# Refreshable Braille Display with Adjustable Cell Size for Learners with Different Tactile Sensitivity

Md Mahmud Hasan Saikot and Kazi Ragib Ishraq Sanim

**Abstract**—Braille is one of the most popular mediums of education for the blind. However, learning braille requires trainers and a lot of practice. Additionally, different individuals have different levels of tactile sensitivity at their fingertips. The tactile components get often overlooked in most braille learning devices and related studies. Our solution is a single cell refreshable braille display with six custom-made electromechanical flapper actuators. It incorporates speech functionalities to facilitate self-learning and independent operation. The cell size can be adjusted according to the learner's preference by moving the actuators. The device can provide standard braille cell dimensions and elevation as well. It is designed to help learners with different tactile perceptions improve themselves through practice and adapt to standard size braille. The operational conditions and force analysis of the braille dots were performed. Two tests were also performed with two different cell sizes to evaluate the device with several blind students. The device is very affordable and easy to maintain. It can also be used to teach braille to the sighted.

**Index Terms**—Tactile display, Electromechanical actuator, Haptic interface, Tactile sensitivity, Braille learning

## 1 Introduction

Our society is progressing with the advancement of technology. Unfortunately, most of these technological advancements and knowledge are enjoyed by people with vision. Visually impaired people are left in the dark literally and figuratively. The available technology is either not compatible or too out-of-reach for them to avail. They read and write with the help of braille. Braille is a form of written system for visually impaired people in which characters are represented by patterns of raised dots that are felt with the fingertips [1]. It was invented by Louis Braille [2]. Conventionally, blind people need to learn braille through a tedious process of memorizing the pattern of dots for every letter of the alphabet. Specialized schools are necessary for this process. According to a report of 2010, only 10% of the people with visual impairments in Canada were able to read braille [3]. Most blind children in South Africa cannot afford specialized schools [4]. They are perceived as a burden to society and are instead kept at home. Learning to read braille can mitigate this problem by creating higher employment opportunities and financial self-sufficiency [5]. There has been some effort in teaching braille with different methods and devices. Lopez *et al.* identified the requirements of a braille teaching device from the end-user inputs [6]. Their work presented a kano model visualizing the device requirements. The must-haves for the device were availability, affordability, tactile learning, and multi-lingual features. Some of the other features that increased customer satisfaction were ease of use, portability, longevity, descriptive speech integration, etc.

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Apps and games have been developed for teaching braille as well. They eliminate the dependency on trainers. Some of the apps and games that serve the purpose are VBGhost [7], mBRAILLE [8], and Br'Eye [9]. These applications use vibrational feedback and audio instructions to teach the braille alphabets, i.e., the users interact using the smartphone touchscreen and get audio and vibrational feedback. However, they fail to introduce the pupil to the actual tactile perception of braille dots. OntoBraille with RFID tags [10], Electronic Braille Blocks [11], The E-Braille [12], and The Read Read [13] introduce learning by tactile sensing of braille dots with simultaneous listening. The main shortcoming of these devices is that they use a block for each character. This makes the device hard to carry and difficult to manage. All these previous works tend to overlook the different levels of tactile perception of the learners. Studies have shown that the tactile perception of the congenitally blind is better than those who lost their vision later in life [14]. Also, tactile sensitivity may differ with age and experience [15]. The new learners, especially the children, need to be introduced to tactile sensing beforehand. In reality, the standard braille cell which has a 2.3 – 2.5 mm gap between two adjacent dots with a dot base diameter of 1.3 – 1.6 mm [16], is rarely the ideal size for learning. Different cell and dot spacing are needed for different individuals at the beginning until they gradually get accustomed to the standard size. For new and tactually lesser sensitive learners, larger dot spacing is more suitable, while those with good tactile sensitivity could learn using the standard size cell. Therefore, most learners would require different cell size throughout the whole learning process.

A single character refreshable display with increasing or decreasing dot spacing and dot diameter while maintaining the aspect ratio would prove very useful to eliminate all the drawbacks discussed to this point. The device would use phonic feedback to teach braille and help the users operate

it independently. As it would also achieve the standard size, the learners could use it for occasional short reading too; enabling them to practice. This way, the learner would not have to get accustomed to new systems/devices every time he/she decides to change to a new cell size.

Tactile displays are made using different types of actuators. The choice of the actuator depends on the bandwidth, resolution, displacements, and force generated by the display [17]. Braille displays for learning usually require a moderate bandwidth with a certain range of forces and displacements. Previously, a lot of braille displays have been made using stepper motors [18] [19], piezoelectric actuators [20] [21], pneumatic actuators [22], actuators based on SMA wires [23], electroactive polymer actuators [24], solenoid based actuators [25] – [27], electromechanical relay based actuators [28] – [31], and flapper actuator [32]. The piezoelectric actuators are very fast and are used by a lot of braille displays. However, they require high voltage (100-300V) to operate, and the fabricated products are generally quite expensive [33]. Pneumatic actuators are inconveniently bulky and slow [34] [35]. The SMA wire-based actuators get significantly hot while operating and need heat dissipation [23]. They are also difficult to control due to hysteresis [36]. The polymer actuators are challenging to fabricate, thus costly. Not to mention the wear, tear, and the right amount of deflection of the dots are also concerning. Furthermore, most of the devices discussed until now have non-standard and larger cell sizes. It is important to familiarize the users with the standard braille. The device by Bettelani *et al*. [25] achieved the standard size, yet they used solenoids that increased the cost of the display. The electromechanical relay or the flapper actuator could be an inexpensive and energy-efficient solution. However, the existing works that use relays as actuators have some limitations. The relay setup of Agam [28], Ranjith *et al.* [29], and Ignat *et al.* [30] do not achieve standard braille cell dimensions. The braille cell of Gayathri *et al.* [31] does not provide uniform deflection of the braille dots. Furthermore, these studies in [27] – [31] do not perform force analysis of the braille dots. Bettelani *et al.* [25] provided a force analysis in their work. However, all the works in [25], [28] – [31] energize the actuators to lower the braille dots from a normally raised position, i.e., a braille dot will lower when the corresponding relay or solenoid is activated. This type of actuation has certain disadvantages. Since the raised dots are not energized, they may have little resistance and move downwards when the reader applies force on the dots. Indeed, Bettelani *et al.* [25] reported that the raised braille dots in their device have a displacement in the range [0.06, 0.2]mm depending on the force from the fingertip. Since the force on all the dots may not be the same, this type of actuation could provide a non-uniform reading experience. This is also true for the devices that are made with relays [28] – [31] since the flapper of the relays (Fig. 3) have very small resistance to movements when the relay is not activated. One way to counter this non-uniform actuation of the braille dots is to incorporate some type of hard end stops [37] or use energized relays for the raised dots. Only the BrailleCursor designed by Loconsole *et al.* [26] used hard end stops. But the design of BrailleCursor was not intended for learning braille. The device of Adnan *et al.* [27] uses energized solenoids to raise the braille dots; however, the standard braille cell size and dot elevation are not achieved. The setup of Yeh and Liang [32] is not suitable for achieving variable dot spacing. The display of Motto Ros *et al.* had the feature of reconfigurable braille [38]. Unfortunately, it was not aimed at learning, and the 8x8 matrix of piezoelectric actuators increased the cost of production and operation.

This study aims to solve the problems mentioned before. The objective of this study is to develop an innovative, portable, and affordable refreshable braille display, convenient for various learners with the unique mechanism of adjustable braille cell sizes. It would help the visually impaired learn braille without requiring additional trainers and make the process much more intuitive than the existing system.

The device exploited the flapping mechanism of six modified electromechanical relay switches to actuate the dots of a braille character. To meet the size constraints of a standard braille character, i.e., to actuate six dots within a fingertip, the actuators were placed in a two-story 3D printed structure (Fig. 5). The structure was specially designed to allow the dot spacing to be adjustable. Due to the low cost of relay switches, the unit became very affordable. The cost analysis is shown in Table 2. The cost of the display was under 10$ (cost of 6 relays and 3D printing). A high refresh rate was also achieved due to the low response time of the relays.

### 1.1 Learning braille

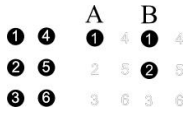



Fig. 1. A braille cell with dot positions numbered.

The numbering of the dot positions in a braille cell is shown in Fig. 1. Characters A and B are also shown. Braille learning is a rigorous process. First, the knowledge of alphabets is essential. For sighted children, it takes practice to memorize the sequences of the alphabet and their appearances. The process is a lot more difficult for visually impaired children. They need to learn the corresponding dot positions in addition to the letters of the alphabet. For example – B is 1, 2 (Fig. 1). The children also go through developing tactile and spatial senses; learning to read braille starts after this. For identifying a letter by its corresponding braille dots, they must be able to identify the positions of the dots raised. Not to mention, even a tiny mistake in spatial perception could make a letter appear entirely different to a blind person. Those with a good tactile perception can easily grasp the concept of braille with standard braille cells. Those who do not, however, need bigger-sized braille like Jumbo American Dot, Giant English Dot, etc. shown in Fig. 2 [16]. After this, they proceed on to practice tracking multiple characters.

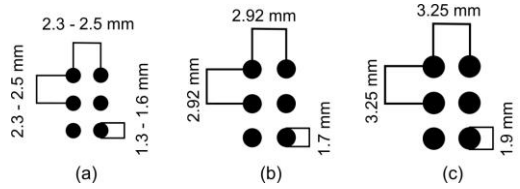



Fig. 2. Different braille cell sizes [16]: (a) Standard size (b) Jumbo American (c) English giant dot.

## 2 DEVICE DESIGN

The device has two main units: a display unit and a processing unit. Fig. 15 shows the working prototype of the device. The processing unit took in user inputs through buttons and controlled the display accordingly. The braille characters were shown on the mechanical display. The mechanism for making dot spacing variable is discussed elaborately throughout section 2.1. and the method of changing dot diameter is discussed in section 2.1.2.4.

### 2.1 The display unit

The display unit was compactly designed to achieve the standard braille cell size. Its dimensions came to be 5cm x 5cm x 4cm after fabrication. The fabricated display unit is shown in Fig. 13a and 13b. It was made with a 3D-printed housing designed to support six custom-made actuators.

#### *2.1.1 Actuators*

The display had to be fast but also reliable while keeping the energy consumption and cost to the minimum. A custom actuating mechanism was devised as the solution, which used the flapping mechanism of electromechanical relay switches. Six miniature 6V relays of dimension 15x10x10 mm were used [39]. Fig. 3a shows the inside of a relay. Inside, there is some gap between the movable and stationary contact points. A constraint plate limits the movable point to move any further than 1 mm. A deflection of about 2 mm was obtained by removing the constraint plate. The deflection was transferred to the braille dots through actuator pins made of steel wires. One end of the wire was soldered onto the movable contact point, which is depicted in Fig. 3b. It transferred the deflection to the tip of the pin/wire when the relay was activated. An activated relay with the pin is shown in Fig. 3c.

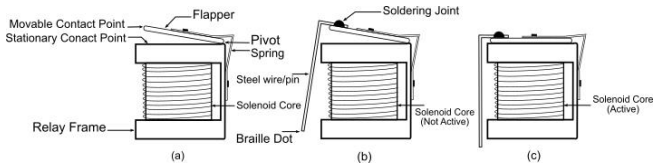



Fig. 3. Electromechanical relay diagram: (a) Inactive relay. (b) Relay flapper attached with the pin. (c) Activated relay

The arrangements of the relays within the device required the steel wire to be bent into two distinctive shapes – a 3D "double L" and a 2D "L" shape. The "double L" shape is shown in Fig. 4a. This type of pin was required for the relays actuating 1, 3, 4, 6 position dots. The remaining 2, 5 position dots required the "L" shaped pins. Both shapes are discussed adequately later in the section 2.1.2.1 and 2.1.2.2.

It was observed that the deflection of the tip of the pin was not completely vertical. There was also a horizontal component of the motion as the flapper was rotating around a pivot. So, the length of the pin was kept as short as possible to reduce the horizontal component. The flapper had a return mechanism using a cantilever spring. When the relay was powered on, the armature/flapper adhered to the core. But when it was turned off, the spring restored the armature to its original position. A little piece of the outer casing of the relay was removed, as shown in Fig. 4b., just so the tip of the flapper was uncovered. Thus, the forms of the relays were unchanged, and they fitted well in the assembly. The pin was then soldered to the flapper, which made it ready to be used as an actuator. The actuator for 1, 3, 4, 6 dots is shown in Fig. 4c.

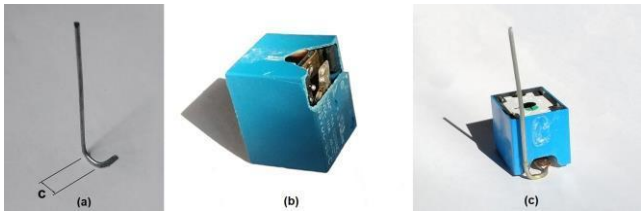



Fig. 4. Double L shaped pin for 1,3,4,6 dots. (b) Hacked relay with exposed movable contact point. (c) Actuator for 1,3,4,6 dots. The pin is attached to the relay.

#### *2.1.2 3D printed housing*

The housing was designed to be two-storied, as shown in Fig. 5, to accommodate six relays in a small space. Four relays were placed on the top floor in an optimum orientation. The other two relays were placed on the bottom floor. All the relays were held in position by special guides shown in Fig. 5c. These guides had several rack-like teeth added on the inner side. The relays had a longitudinal bit added on both sides to allow them to slide inward or outward in steps and lock in position. This enabled the user to change the inter-dot spacing of the display.

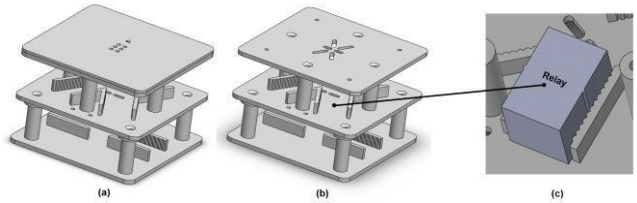



Fig. 5. 3D printed housing (relays are not shown): (a) With top cover. (b) Without top cover. (c) Relay held in guide with rack teeth. The relay has a longitudinal bit to mesh with the guide.

The relays were positioned upside down on the floor of the housing. The floor prevented the movable contact point of the relay from going down more than they were supposed to. The housing assembly had four parts – the bottom layer, mid-layer, top layer, and top cover.

2.1.2.1 Bottom layer: The bottom layer had two relays with longer pins on them. Two guides held the relays in position. They are shown in Fig. 6a.

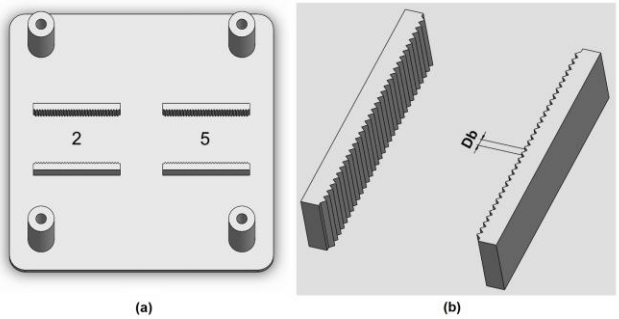



Fig. 6. Bottom layer: (a) Guides for 2 and 5 dot controlling relays. (b) Rack teeth on the guide. Pitch of the rack teeth is Db.

The two relays marked by 2 and 5 controlled the 2, 5 dot positions. They required 2D "L" shaped pins. The pitch of the rack teeth on these guides was $Db = 0.5mm$ (Fig. 6b). It allowed each relay to move in 0.5mm steps. So, the horizontal dot spacing would change in steps of 1mm when both relays are moved one step outwards. There were four columns or screw holes on this layer supporting the mid-layer on top of it.

2.1.2.2 Mid Layer: The dimensions of the dots of a braille cell can be seen in Fig. 7. The horizontal ($Dh$) and vertical ($Dv/2$) dot spacing were equal according to most of the standard braille cell sizes [16]. The present design also implemented $Dh = Dv/2$, making the aspect ratio of the cell $\left(\frac{Dh}{Dv}\right) = \frac{1}{2}$.

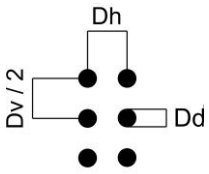



Fig. 7. Braille cell dot dimensions.

The mid-layer contained four relays arranged in an optimal configuration. The configuration is shown in Fig. 8. In this figure, the relays are 15x10mm in dimension and numbered by the dot positions they control, i.e., 1, 3, 4, 6. These braille dots are indicated by four points in the center of Fig. 8. They are offset from the respective relay center by a distance of $c$. This is the reason for using the 3D "double L" shaped pins discussed before (Fig. 4a) in section 2.1.1.

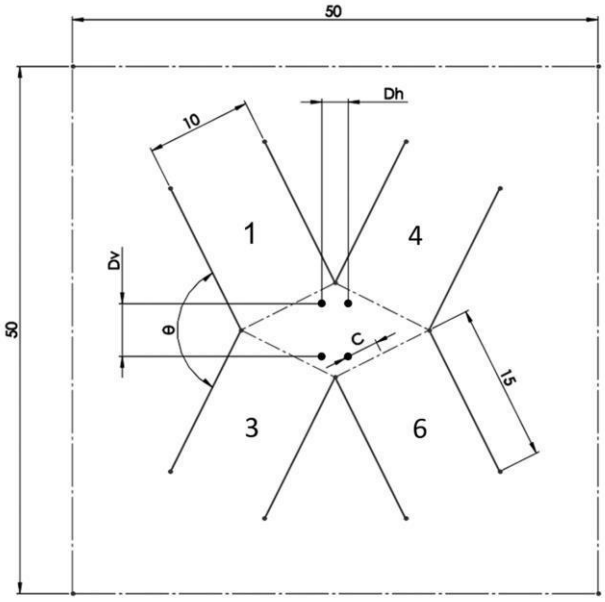



Fig. 8. Relay arrangements in mid layer (all dimensions are in mm). The relays occupy the positions marked by 1, 3, 4, 6.

In Fig. 8.,

$Dh$ = Horizontal dot spacing

$Dv/2$ = Vertical dot spacing

$\theta$ = Obtuse angle between two relays

$c$ = Offset distance of the 1, 3, 4, 6 dots from the center of the respective relays

The horizontal dot spacing ($Dh$) was controlled by two horizontally adjacent relays on this layer and the bottom layer. The vertical dot spacing ($Dv/2$), however, was controlled individually by the 1, 3, 4, 6 position relays. So, the relays in this layer had to vertically move double the distance of horizontal movement to keep the aspect ratio constant. This made the angle in Fig. 8 to be $\theta/2 = arctan\,(2)$. So, the pitch of the rack teeth in the mid-layer guide (Fig. 9b) was, $Dm = Db/cos(\theta/2) = 1.118mm$. Here, $Db$ is the pitch of the rack teeth in the bottom layer guide (Fig. 6b).

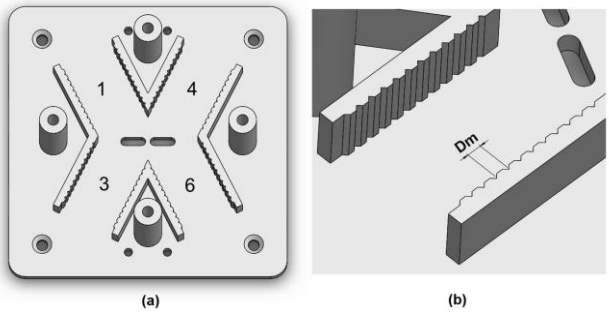



Fig. 9. Mid layer: (a) Guides for 1, 3, 4, 6 relays, and two wide slots in the middle for 2 and 5 pins to come up through. (b) Pitch of the rack teeth on the middle layer is Dm.

The mid-layer is shown in Fig. 9. It had two 2mm wide slots to allow the 2 and 5 actuator pins from the bottom layer to come up through it. This slot also restricted the horizontal dot spacing of positions 2, 5 from extending more than 4mm from the standard size. This layer also had four columns to support the top layer on it.

2.1.2.3 Top layer: The top layer is shown in Fig. 10a. This layer allowed the pins of all the actuators to come up through the slots made on it. The slots only allowed the pins to move from the minimum to the maximum cell size. There were four holes for the screws and four blind holes for mounting the top cover on it.

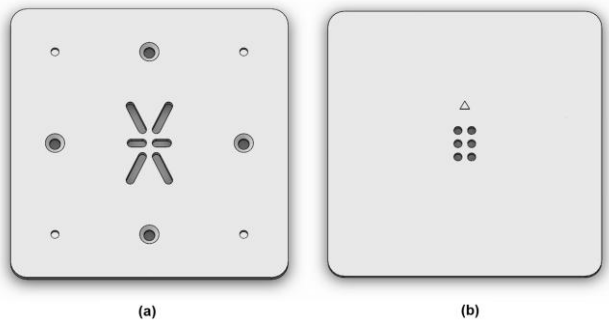



Fig. 10. (a) Top Layer. The slots are for the actuator pins to come up through and move. (b) Top cover, standard cell.

2.1.2.4 Top cover: The top cover is a removable cover that sat on the top layer. This layer had six holes for the braille dots and covered the slots of the top layer to give a smooth reading experience. Different covers were designed for different cell sizes. Each cover had a triangle-shaped protrusion for indicating the up orientation of the display. The top cover with the standard braille cell (2.5x2.5mm) is shown in Fig. 10b.

The dot base diameter is 1.3 – 1.6 mm according to many of the standards [16]. The steel wire selected for the actuator pins was SWG 17. Its diameter is 1.42 mm, which is within this range. So, the tip of the pin was acting as the dot for the standard cell. However, the dot base diameter increases with increasing dot spacing in bigger sizes, i.e., Jumbo American, Giant English Dot (Fig. 2). So, the device must allow the dot diameter to increase with increasing inter-dot distance. This functionality is implemented in the top cover. The top cover for the bigger cell sizes had larger holes for the dots. This created some space around the tip of the pin. The space around the pin is shown in Fig. 11a.

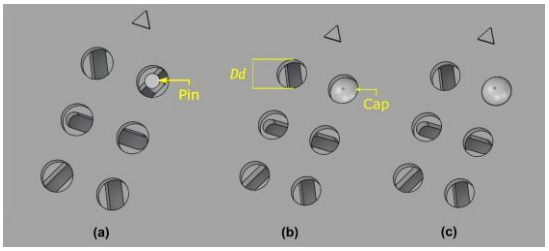



Fig. 11. (a) Top covers for bigger cells have larger holes and create empty space around the actuator pins (only one is shown). (b) The pins are attached with a cap of diameter. (c) The cap acts as a raised braille dot.

The tip of the pin could be attached with little caps of the diameter required for that specific cell size and top cover. The attached cap is shown in Fig. 11b and 11c. Conversely, the caps could be integrated into the top cover using slots that allow them to move freely between the raised and lowered dot positions. This would prove to be convenient when changing the cell size since the users would not have to attach individual caps by themselves. The dot diameter would automatically change when the top cover is changed.

### 2.2 Processing unit

A Raspberry Pi 3b was used as the processing unit for fast prototyping and its ease of use. It is shown in Fig. 15. It has 40 GPIO (General Purpose Input Output) pins. Six output pins were used for controlling six actuators individually. Four more input pins were used for the control switches of the device. The raspberry pi also has wireless LAN (Local Area Network) for connecting to an android/computer. There are several USBs (Universal Serial Bus) to connect to as well. Chip on board (COB) may be used as the processing unit in mass production to bring down the cost of the device.

### 2.3 Electronics

A 5V power bank was used to power the processing unit. The relays were powered by a 12V 350mA output wall adapter. The wall adapter allowed the voltage to be changed. Each relay was switched by an NPN 2N2222 transistor. The transistors enabled supplying the current required by the relays. A custom circuit containing all the transistors was designed. There were four buttons on the device for control, shown in Fig. 15. Their functionality is discussed in section 2.4.

### 2.4 Software

Python was used for developing the program. It took user inputs through the control buttons and controlled the GPIO pins of the raspberry pi to actuate characters on display. The python module eSpeak was used for speech integration with the display.

#### *2.4.1 Voice module*

eSpeak is a speech synthesizer program. It is an English text-to-speech system that speaks out the inputs given to it. The speech integration not only helped the users with the learning process but also allowed them to operate the device more efficiently. The users were given voice directions for each functionality of the device. Voice feedback was also provided for each button press.

#### *2.4.2 General flowchart*

For the sake of conciseness, the general flowchart of the device program is discussed here without getting into the details of the programming methods. The user inputs are a vital part of every program. The four buttons were the user inputs to the program to operate the device. They were named Mode, Next, Previous, and OK according to their functions. The flowchart of the program is shown in Fig. 12. It helps us understand the workflow of the program and the device. The program could switch between modes (learning/reading) triggered by the Mode button at any time. The user is informed by the voice as soon as the device enters a mode.

2.4.2.1 Learning mode: Inside the learning mode, there are two options to choose from – Learn alphabets and Practice alphabets. The users can toggle between the options by pressing the Next/Previous button and confirm by pressing OK.

In Learn alphabets option, the letters of the alphabet are shown from a-z, and each letter stays on the display until the user triggers the next/previous buttons. The program can detect repeated Previous/Next button press without pause to skip the same number of characters. As a character is actuated on the braille display, the letter is spoken out simultaneously. The user could also listen to the displayed letter and corresponding dot positions by pressing the OK button anytime. The dot positions are crucial for new learners with less tactile sensitivity, as they might not be able to identify all the dot positions correctly. They have the option to correct themselves by listening to it and decide if they require a larger cell size.

In the Practice alphabets option, random letters of the alphabet are shown on display, and simultaneous voice is not used, contrary to the Learn alphabets option. The users can, however, press the OK button anytime to listen to what the display is currently showing. This enables them to practice alphabets with the device.

2.4.2.2 Reading mode: There is also a reading mode for reading electronic documents with the device. In reading mode, the program reads a document on the connected USB and sends output characters to the display. The user can navigate through the USB drive or the default files using the Next/Previous buttons. The voice module speaks out the selected file name as the user navigates

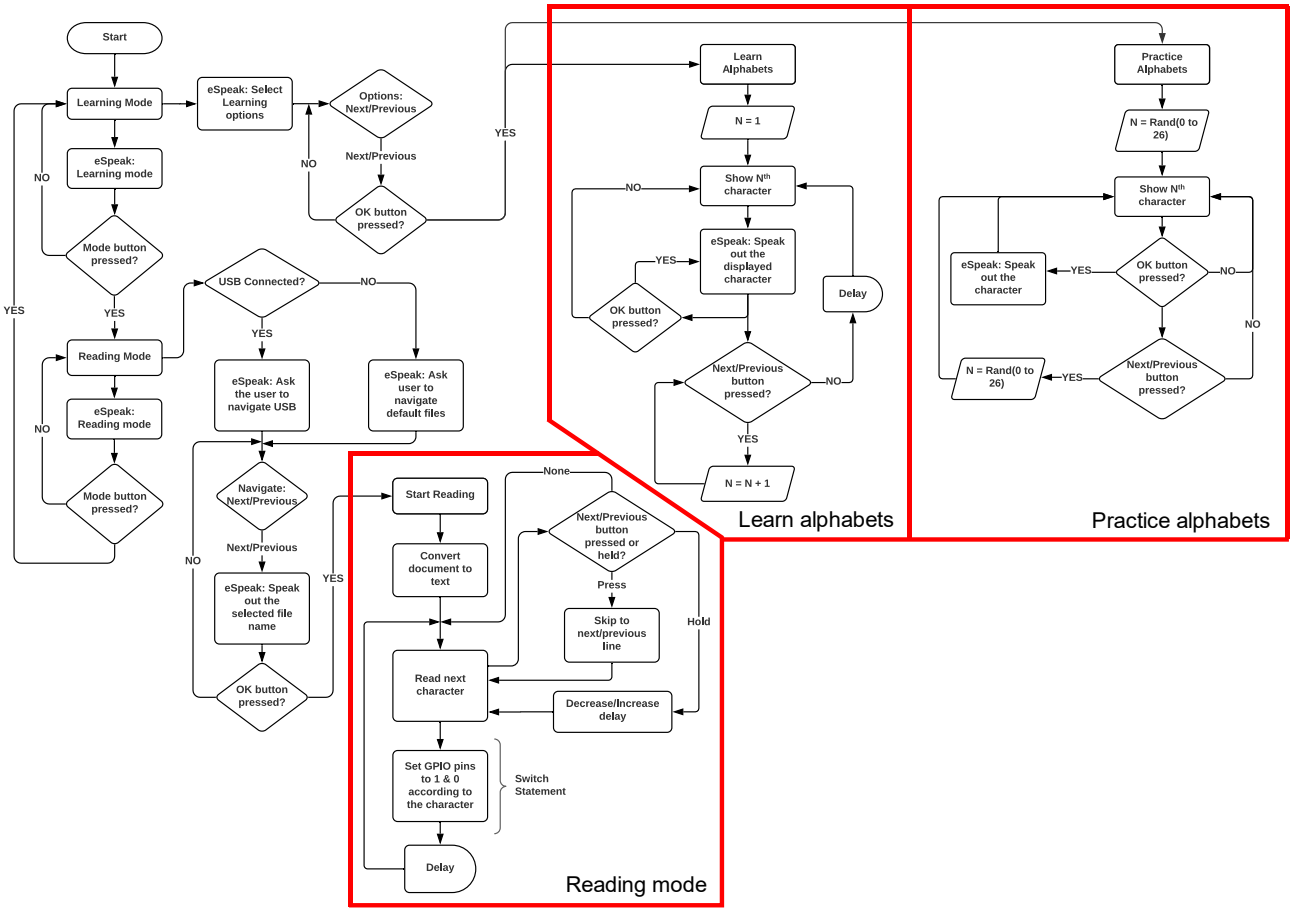



Fig. 12. General flowchart of the device program.

through. A file can be chosen for reading by pressing the OK button. The program calls the Start Reading function with the chosen file.

Firstly, the document is converted to text. After that, one character is extracted at a time, and specific GPIO pins are set high and low according to the character. This is done with the help of a switch statement. After each character, a small amount of delay is given before showing the next character. During this pause, all the GPIO pins were set to low; thus, making all the dots lower. A longer delay was made between the characters for indicating a "space". Lines can be skipped by pressing the Next/Previous buttons. The duration one letter stayed on display is the delay. Holding the Next/Previous buttons changed the delay and made the display fast/slow. This allowed the users to change the refresh rate of the display according to their likings.

## 3 Operating Conditions

It is important to find out the optimum operating conditions for the device. The upward force for comfortable braille reading should be over 15 gf. The force generated by the dots, i.e., the pins of the device, was determined using weights on the pins. The power requirement of the device for producing the required force was determined with the weights on. The voltage and current consumption were monitored for this purpose. The temperature of the relay coil was also determined at the specific powers from the datasheet [39].

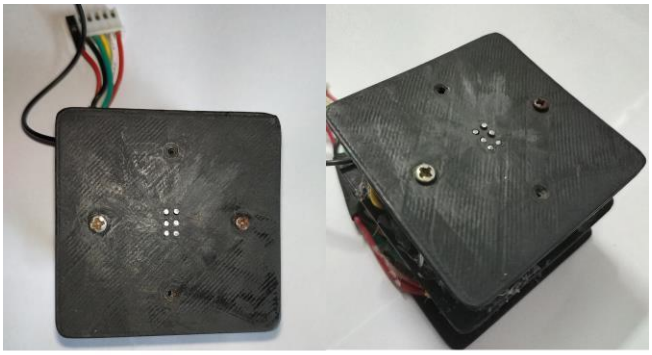



Fig. 13. The display: (a) All dots are lowered. (b) 4, 5 dots are raised.

TABLE 1
Force generated by the relay at different operating conditions.

| Voltage (V) | Current (Amps) | Power (Watts) | Force (gf) |
|---|---|---|---|
| 4.2 | 0.07 | 0.294 | 2 |
| 5.35 | 0.09 | 0.482 | 12.25 |
| 6 | 0.1 | 0.6 | 16.3 |
| 6.75 | 0.11 | 0.743 | 26.5 |

Table 1 presents the experimental data of required voltage and currents for every specific force generated. It can be seen that the voltage should be equal to or greater than 6V to generate the required force for easy reading of the braille dots. At the operating point of 6V, the device draws 0.1 amps of current to generate a 16.3 gf force upward. The power consumption is 0.6W.

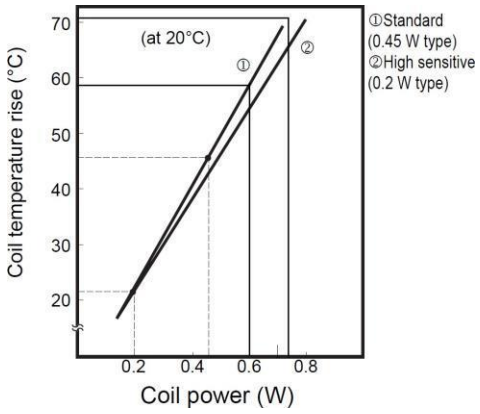



Fig. 14. Coil temperature rise vs Coil power [39]

The coil temperature can be determined using the power consumption from Fig. 14. It shows the data at an operating condition of 20 ◦ C. When the coil power is 0.6W, the coil temperature rise is ∼ 58 ◦ C. This is well within the operating region of the relay. However, if the operating voltage is changed to 6.75 Volts, the power consumption becomes 0.743 Watt and consequently a rise of ∼ 71 ◦ C in the temperature. This temperature is moderately high for the relay, and it is not recommended to operate in such temperatures for a prolonged time. Thus, the operating point is chosen as 6 Volts.

The GPIO pins on the raspberry pi can be switched very fast, thus, has a negligible effect on the speed of the display. The speed of the display was determined by the operate-time (max. 5 ms) and release time (max. 5 ms) of the electromechanical relay [39]. Due to the fast speed of the relays (max. 10 ms), the display speed was fast as well. The refresh rate of the display could go higher than 10 Hz.

The mechanical life of the relays is $5x10^6$ cycles, while the electrical life is $1x10^5$ cycles at 2A/12VDC resistive load [39]. For our purposes, the device does not carry any electrical load. Even after considering a safety factor of 10, the mechanical life of the relays is sufficient for the device to be long-lasting and reliable.

TABLE 2
Cost analysis and comparison with the device of Bettelani et al.

| Items | Bettelani et al. Device [25] | | Present study device | |
|---|---|---|---|---|
| | Quantity | Price ($) | Quantity | Price ($) |
| Arduino micro | 1 | 6.28 | 0 | 0 |
| Rasp. PI zero | 0 | 0 | 1 | 5 |
| Solenoids | 6 | 26 | 0 | 0 |
| Relays | 6 | 8.48 | 6 | 7.2 |
| Battery/Power | 1 | 2 | 1 | 4 |
| Circuits | 0 | 0 | 1 | 3 |
| Total cost | | 42.74$ | | 19.2$ |

Table 2 shows the cost analysis of the present device components and compares it to a similar device by Bettelani *et al* [25]. It is evident the present study offers a more cost-effective solution compared to the device of Bettelani *et al.* Note that we have shown the price of Raspberry PI zero instead of PI 3B used in our prototype. PI zero provides the same functionality for our device, and it is better suited for cost-effectiveness. The present device cuts down a lot of the costs using relays as actuators instead of solenoids.

## 4 Experiment and Discussion

The device was tested on seven male students of the Blind Education and Rehabilitation Development Organization (BERDO) in Bangladesh. All the subjects for the experiment had to be completely blind who are in the learning phase and know the English braille alphabet, i.e., the dot positions for the English alphabet. Only seven subjects were chosen due to the paucity of students who met the criteria. BERDO provided informed consent on behalf of all of them. The participants had used the device only once before the experiment and were familiar with the display.

### 4.1 Protocols of the experiment

The device was placed on a table and the participants were asked to sit on a chair one by one to use the device while we sat on the opposite side to instruct the participants and collect data. Two tests were performed: one with standard size (2.5x2.5 mm) and another with a larger cell size (3.5x3.5 mm). The test with the standard cell size was performed on all seven participants first, before moving on to the test with the larger cell size. For both tests, 10 trials were performed for each of the participants. In each trial, a letter was raised on the device display from a pre-selected set of 10 letters (b,e,h,j,o,r,s,t,v,y). This set of letters was chosen considering the varying number of dots raised per each letter. These letters were raised on the display as braille dots and the participants were allowed as much time as they needed for identifying each letter. They were asked to speak out the letter once they had identified it. The letters were raised manually by us to keep the testing condition constant for every participant. This also made the duration of the experiment shorter since the participants did not have to learn to operate the device. After the test with the standard size, the participants that did not perform well with this size were asked to try on a larger cell. The purpose of the second test was to observe if these participants could improve their accuracy on a larger cell size. To meet the purpose, one larger cell size was enough. The size of the larger cell was chosen by us to ensure all the participants for the second test have the same testing conditions.

### 4.2 Results

Table 3 shows the test done with the standard cell size. From Table 3, we can see that the more experienced subjects 1, 2, and 3 had high accuracy with the standard size cell. This suggests the device was highly readable.

It is also observed that subjects 5, 6, and 7 had low accuracy, and they required more time to identify the letters correctly. This could be due to two reasons – low tactile sensitivity or lack of experience. However, moving to a larger cell size after the standard size increased their performance significantly. Table 4 shows the test done with the larger cell size (3.5x3.5 mm).

TABLE 3
Test with the standard size (2.5x2.5 mm) cell

| Sub-ject | Age (years) | Acquainted to braille for (years) | Letters shown | Letters correctly identified | Accuracy (2.5x2.5 mm) (%) | Average time required (s) |
|---|---|---|---|---|---|---|
| 1 | 17 | 3 | 10 | 10 | 100 | 2 |
| 2 | 14 | 2 | 10 | 9 | 90 | 2 |
| 3 | 13 | 2 | 10 | 10 | 100 | 2.5 |
| 4 | 13 | 0.5 | 10 | 9 | 90 | 3 |
| 5 | 14 | 0.5 | 10 | 6 | 60 | 4 |
| 6 | 13 | 0.5 | 10 | 3 | 30 | 4.5 |
| 7 | 14 | 0.5 | 10 | 2 | 20 | 7 |

For the test presented in Table 4, the participants 5, 6, and 7 were evaluated with a bigger cell size (3.5x3.5 mm) since their accuracy was relatively lower using the standard cell (table 3).

TABLE 4
Test with 3.5x3.5 mm cell

| Sub-ject | Age (years) | Acquainted to braille for (years) | Letters shown | Letters correctly identified | Accuracy (2.5x2.5 mm) (%) | Average time required (s) |
|---|---|---|---|---|---|---|
| 5 | 14 | 0.5 | 10 | 8 | 80 | 2.5 |
| 6 | 13 | 0.5 | 10 | 6 | 60 | 3.5 |
| 7 | 14 | 0.5 | 10 | 8 | 80 | 4 |

These two tests show that different individuals performed best in different cell sizes. The device could be helpful for different individuals to learn braille in different sizes effectively. After the tests, the participants were asked to rate their experience of reading the letters with the device into one of the three categories: good, medium, and bad. Six of them chose good and participant 7 chose medium since he had the most difficulty with the first test with the standard size. One issue some participants seemed to complain about was the standard cell (2.5x2.5 mm) dots being a little rough. The surface finish of the steel pins was not very smooth. The problem could be avoided by giving the pins a better surface finish or machining them using precise methods.

### 4.3 Discussion

A blind user should be able to operate the device without help after getting an introduction to the device tactually. The device allows the user to learn and practice independently. However, they will require detailed instructions on how to change the cell size. For changing the cell size, the top cover has to be removed first. Then each of the relays must be moved equal steps manually inward or outward. This will change the dot spacing. Finally, a suitable top cover corresponding to the cell size has to be placed on top, which will change the dot base diameter. If a blind person faces difficulty changing cell size, he or she could get help from the voice module. When changing the cell size for the first time, they could also learn the

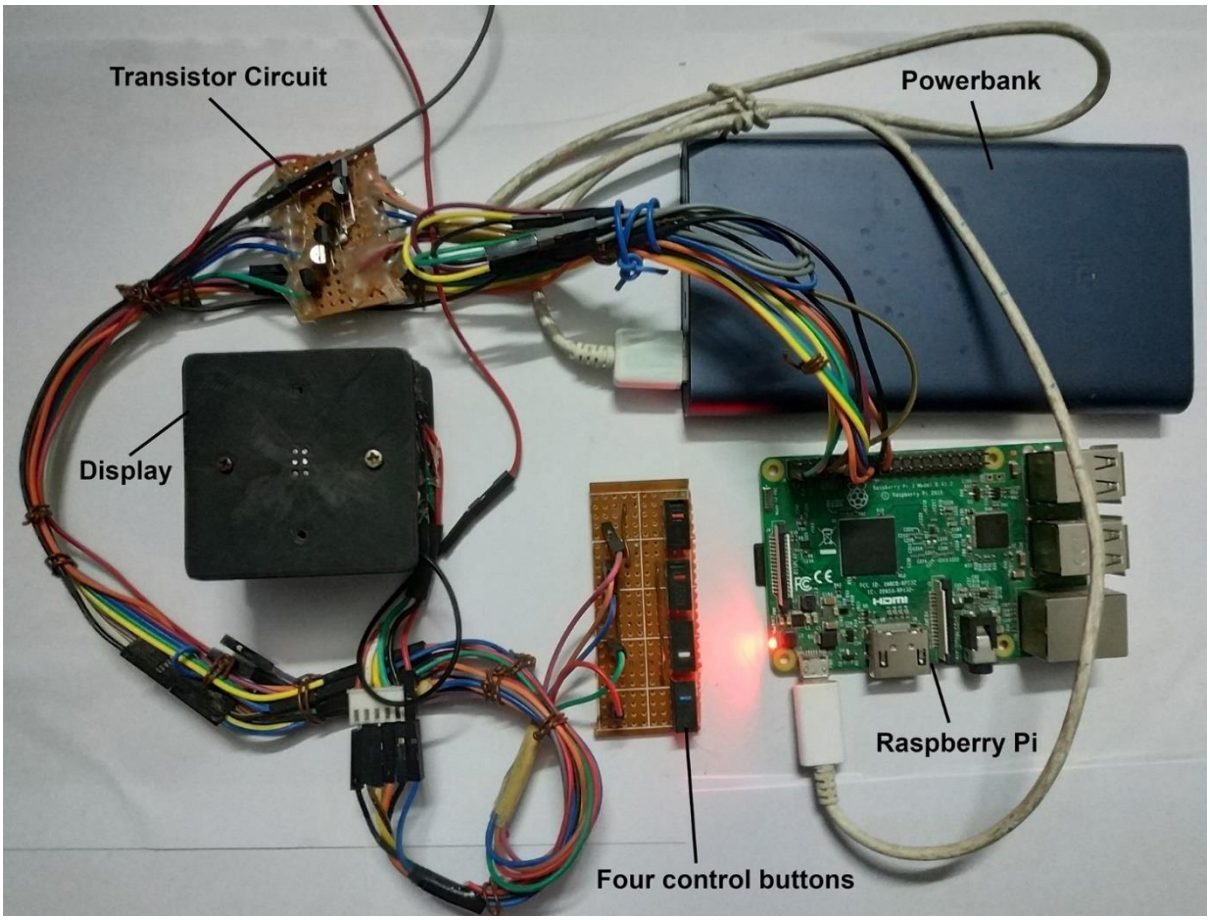



Fig. 15. The working prototype of the device. Raspberry pi is the processing unit powered by the power bank. There are four control buttons and a transistor circuit. The braille dots are actuated on the display.

procedure from a sighted person. The process could be made easier by building the caps into the top covers and making the cover latch magnetically. A sound could also be incorporated when each relay is moved.

In this study, we discussed a design where both the horizontal and vertical dot spacing changed in 1 mm steps. Other cell sizes could be achieved by changing the pitch of the rack teeth of the guides on the bottom layer and mid-layer, i.e., dimensions *Db* and *Dm*. The control buttons should be made intuitive in shape to aid the blind user to differentiate and identify the function by touching.

The device has a reading functionality built into its program. Thus, it can be used for extended reading as well. Ignat *et al.* suggested that reading on a single-cell tactile display has the potential to increase the reading speed of the user [30]. However, tracking is not possible with single character displays, which is necessary for the blinds to read traditionally. The device is best suited for learning the braille combinations, and it would help the learners develop better tactile perception to move on to standard-sized braille. This study agrees with the previous studies and provides a better solution for learning braille.

## 5 Conclusion

In this study, we focused on developing a braille display with adjustable cell sizes for learners with different tactile sensitivity. To the best knowledge of the authors, no work has been attempted to make a refreshable display with adjustable cell size before. This device will help learners to start braille with their preferred size and then move on to standard braille eventually. It also enables them to practice braille, thus improving their tactile perception. It is very affordable (~20$) and portable. Additionally, the device could allow blind students to learn braille independently and remove a lot of burden from the teachers. It would naturalize the process of learning braille using phonetic feedback with tactile stimulation. However, further user studies would be required to fully evaluate the learning performance with the device.

During this study, we found out that relays could be an excellent option for actuating short distances. We came to know some blinds have less spatial sense. We also learned some unique methods used to teach braille. For example, the children who have trouble figuring out the relative positions of the dots are taught by dummy braille dots beside the actual braille dots. A top cover, according to the example, is shown in Fig. 16.

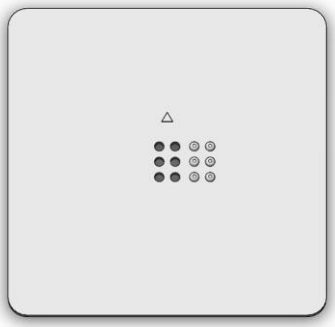


Fig. 16. Top cover, standard cell with dummy dots.

Different top covers could be designed and tested to observe their effect on learning. This could be an excellent direction for future studies. An intelligent app could be incorporated for collecting user data and make the learning process personalized for the student. The app could also be equipped with an OCR (Optical Character Recognition) to convert handwritten texts to braille texts on display. Automatic cell size changing functionality could be employed for ease of use. Voice recognition would be another useful addition. These are some of the directions for further improvements.

## Acknowledgments

We thank our fellow team member Shaumik Rahman Ayon from 'Team Tesseract', who helped with the fabrication of the device and provided great insights that helped the research.

We would also like to show our gratitude to Mr. Sher Ali, Teacher, BERDO, for helping us conduct the experiments with the students and sharing his wisdom with us. Special thanks to Nazim and all the blind students who participated in the tests.

We are also immensely grateful to Md. Faiyaz Jamil for his assistance with some figures and proofreading. We also thank Meril Moni and Muhaimin Mareum Khan for proofreading and commenting on the manuscript.

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